Voltage Regulation

Range: 2.2 – 6.0V

5V
C2 10UF-0603-25V-(+80/-20%)
C13 0.1UF-0603-25V-(+80/-20%)
C1 0.1UF-0603-25V-(+80/-20%)
JUMPER-SMT_2_NO_SILK
R5 100KOHM-0603-1/10W-1%
U5 AP7361C-33FGELDFN-8 (IN, OUT, GND, EN)
C9 10UF-0603-25V-(+80/-20%)
C10 0.01UF-0603-25V-10%
C11 0.22UF-0603-25V-10%
3.3V
MEAS0
JUMPER-COMBO_2_NC_TRACE
D1 100KOHM-0603-1/10W-1%
3.3V_EN
Close this jumper to control the VREG in super low-power modes

RTC

3.3V
1A/3.3V/620mV 0.65uA Leakage 3.3V drop at 10mA
R8 330KOHM-0603-1/10W-1%
RTC_3V
D6 3A/10V/280mV 300uA Leakage
BT1 ML414H_IV01E_BATTERY
1mAhr Capacity

This is the penny-in-the-fuse jumper. Close only if you know what you're doing.

V_USB
BYPO
5V
F2 PPTC_6V2A

LEDs

5V
JUMPER-SMT_2_NC_TRACE_SILK
R20 1KOHM-0603-1/10W-1%
D11 RED
3.3V
R11 1KOHM-0603-1/10W-1%
D2 RED

Cut these jumpers to remove power LEDs.

µSD

3.3V
J11 MICRO-SD-SDIOPUSH-PUSH
VCC, DAT3/CS, DAT2/NC, DAT1/NC, DAT0/SDO, CLK/SCK, CMD/SDI, SHIELD, GND
SD_CS, SPI_CIPO, SPI_SCK, SPI_COPI
C46 0.1UF-0603-25V-(+80/-20%)

BME280 – Temp, Pressure, Humidity

3.3V
C4 0.1UF-0603-25V-(+80/-20%)
C3 0.1UF-0603-25V-(+80/-20%)
U1 BME280 (VDD, VDDIO, GND, CS, SDI/SDA, SCK/SCL, SDO/ADR)
R21 10KOHM-0603-1/10W-1%
I2C Mode
R7 10KOHM-0603-1/10W-1%
Default address: 0x77

UV Sensor

3.3V
JUMPER-SMT_2_NC_TRACE_SILK
Cut this jumper if you want to remove from power add an external UV sensor.
C5 0.1UF-0603-25V-(+80/-20%)
VDD, GND, SDA, SCL (I2C_SDA, I2C_SCL)
I2C address: 0x10

Soil Moisture Sensor Connector

SOIL_MOIST_SIG
SOIL_MOIST_PWR
J4

Lightning Detector

INP
L1 MA5532-AE
C8 1nF
R6 10KOHM-0603-1/10W-1%
3.3V
U3 AS3935QFN
Select SPI mode
VDD, VREG, EN_VREG, AC_GND, GND, GND_PAD
SI, MOSI/I2CD, MISO, SCL/I2CL, CS, ADD0, ADD1, NC, IRQ, INP, INN
SPI_COPI, SPI_CIPO, SPI_SCK, LIGHTNING_CS, LIGHTNING_INT, INP
C6 1.0UF-0603-16V-10%
C7 10UF-0603-6.3V-20%
R19 100KOHM-0603-1/10W-1%

USB

Range: 4.2 – 5.2V

V_USB
J1 USB-C (VBUS, D+, D-, CC1, CC2, SHLD, GND)
USB_D+, USB_D-
R3 5.1k
R2 5.1k

JTAG

3.3V
VCC, GND, RESET, SWDIO, SWDCLK, SWO

Universal MicroMod Connector

3.3V
RESET, BOOT, 3.3V_EN, RTC_3V, VIN, USB_D-, USB_D+, USB_VIN, USB_D-, USB_D+, USBHOST_D-, USBHOST_D+, CAN-TX, CAN-RX, SWDIO, SWDCK, AUD_MCLK, AUD_OUT/CAM_MCLK, AUD_IN/CAM_PCLK, AUD_LRCLK, AUD_BCLK, I2C_SCL, I2C_SDA, I2C_INT, I2C_SCL1, I2C_SDA1, BATT_VIN/3, GND
SPI_SCK1/SDIO_CLK, SPI_SDO1/SDIO_CMD, SPI_SDI1/SDIO_DATA0, SDIO_DATA1, SDIO_DATA2, SPI_CS1/SDIO_DATA3, SPI_SCK, SPI_SDO, SPI_SDI, SPI_CS, A0, A1, PWM0, PWM1, D0, D1/CAM_TRIG, TX1, RX1, RTS1, CTS1, TX2, RX2, G0/BUS0, G1/BUS1, G2/BUS2, G3/BUS3, G4/BUS4, G5/BUS5, G6/BUS6, G7/BUS7, G8, G9/ADC_D-/CAM_HSYNC, G10/ADC_D+/CAM_VSYNC, G11
R12 100KOHM-0603-1/10W-1% 0.1UF-0603-25V-(+80/-20%)
S1 MOMENTARY-SWITCH-SPST-SMD-5.2-TALL-REDUNDANT
S2 MOMENTARY-SWITCH-SPST-SMD-5.2-TALL-REDUNDANT
R10 100KOHM-0603-1/10W-1%
R9 100KOHM-0603-1/10W-1%

Rain

3.3V
RJ11-6 J7
R15 43KOHM-0603-1/10W-1%
D3 DIODE-BAS16J
D4 DIODE-BAS16J
C14 0.1UF-0603-25V-(+80/-20%)
RAIN

Wind

RJ11-6 J13
R16 43KOHM-0603-1/10W-1%
D5 DIODE-BAS16J
D8 DIODE-BAS16J
D9 DIODE-BAS16J
D10 DIODE-BAS16J
C12 0.1UF-0603-25V-(+80/-20%)
R14 10KOHM-0603-1/10W-1%
WDIR, WSPEED

GPIOs

J5 (SPI_SCK, SPI_COPI, SPI_CIPO, HEADER_CS)
J9 (PWM0, PWM1)
J8
J6 (TX1, RX1)
J12 (TX2, RX2)
SWDIO

Qwiic

3.3V
JP1 JUMPER-SMT_3_2-NC_TRACE_SILK
Cut this jumper to disable pull-ups on I2C bus.
R4 2.2KOHM-0603-1/10W-1%
R1 2.2KOHM-0603-1/10W-1%
I2C_SCL, I2C_SDA
J3 QWIIC_RIGHT_ANGLE (SCL, SDA, VCC, GND)
J14 (SCL, SDA, VCC, GND)

sparkfun
qwiic
open hardware

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TITLE: ${PROJECTNAME} – Carrier
Design by: >DESIGNER Priyanka Makin
REV: >REV V14
Date: ${CURRENT_DATE}
Sheet: ${#}/${##}

Sheet: /
File: working.kicad_sch
Title:
Size: User
Date:
Rev:
KiCad E.D.A. eeschema 7.0.6
Id: 1/1